

Ex. No. : 02
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A PYTHON PROGRAM TO IMPLEMENT LINEAR REGRESSION

Aim:

To develop a Python program to implement Linear Regression and analyze the relationship between dependent and independent variables.

Requirements:

Jupyter Notebook with Python, Corona Virus Report dataset (country_wise_latest.csv), and required Python libraries (pandas, numpy, matplotlib, sklearn).

PROGRAM:

Linear Regression

```
import kagglehub
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

path = kagglehub.dataset_download("imdevskp/corona-virus-report")
df = pd.read_csv(os.path.join(path, "country_wise_latest.csv"))

X = df[['Confirmed']]
```

```
y = df['Deaths']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
random_state=42)
```

```
model = LinearRegression()  
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Intercept:", model.intercept_)  
print("Coefficient:", model.coef_[0])  
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))  
print("R2 Score:", r2_score(y_test, y_pred))
```

```
plt.scatter(X_test, y_test)  
plt.plot(X_test, y_pred)  
plt.xlabel("Confirmed Cases")  
plt.ylabel("Deaths")  
plt.title("Linear Regression - Confirmed vs Deaths")  
plt.show()
```

Output

:



Result:

The Linear Regression model shows a strong positive relationship between confirmed cases and deaths.

The R^2 score is close to 0.90+, indicating that the model predicts deaths accurately based on confirmed cases.